

CAPE CAMPBELL, New Zealand

WMO: 935970

Lat: 41.7264S

Lon: 174.2761E

Elev: 6

StdP: 101.25

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	6.0	-2.4	3.1	7.2	-0.7	3.6	7.7	18.6	10.3	17.4	10.4	6.9	200	0.822

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.6	20.4	16.0	19.5	15.9	18.9	15.7	17.3	19.0	16.8	18.5	16.4	18.1	10.8	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	11.7	18.3	16.0	11.4	17.9	15.5	11.0	17.6	48.8	19.2	47.3	18.7	46.0	18.2	19.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.5	15.7	14.5	DB	3.8	23.2	0.8	1.3	3.3	24.2	2.8	25.0	2.4	25.7	1.9	26.7	(4)
(5)			WB	1.2	18.2	0.9	0.7	0.5	18.7	0.0	19.1	-0.5	19.5	-1.2	20.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	16.2	16.3	15.7	13.8	12.2	10.2	9.4	9.8	10.8	11.9	13.3	15.1	(6)	
	DBStd	2.94	1.60	1.43	1.65	1.67	1.80	1.73	1.60	1.58	1.72	1.74	1.79	1.71	(7)	
	HDD10.0	89	0	0	0	1	3	18	31	23	11	4	1	0	(8)	
	HDD18.3	1996	69	58	84	136	189	243	277	264	224	199	152	101	(9)	
	CDD10.0	1139	192	177	175	115	72	24	12	17	36	62	98	158	(10)	
	CDD18.3	5	2	2	1	0	0	0	0	0	0	0	0	1	(11)	
	CDH23.3	1	1	0	0	0	0	0	0	0	0	0	0	1	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	7.8	8.1	7.5	7.8	7.3	7.5	7.3	7.3	7.2	8.0	8.4	8.5	8.2	(14)	
(15) Precipitation	PrecAvg	654	51	41	46	54	58	73	74	62	53	52	50	46	(15)	
	PrecMax	1061	106	136	125	147	167	153	197	166	114	116	129	101	(16)	
	PrecMin	371	0	0	3	5	13	21	16	11	0	5	6	5	(17)	
	PrecStd	163	28	34	27	38	35	35	44	35	32	29	33	27	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	21.5	21.1	18.9	17.4	15.3	14.1	14.5	16.2	17.6	19.7	21.0	(19)	
		MCWB	16.1	16.5	15.5	14.5	13.3	11.8	10.9	10.6	11.6	12.4	13.4	14.7	(20)	
	2%	DB	20.2	20.1	19.6	17.6	16.3	14.3	13.2	13.3	14.8	16.1	17.6	19.3	(21)	
		MCWB	16.4	16.6	15.6	14.2	13.0	11.4	10.4	10.3	11.2	12.5	13.4	15.3	(22)	
	5%	DB	19.3	19.3	18.8	16.9	15.5	13.6	12.5	12.6	14.0	15.2	16.6	18.3	(23)	
		MCWB	16.1	16.3	15.5	14.0	12.7	11.1	10.0	10.0	11.1	12.2	13.2	15.0	(24)	
	10%	DB	18.6	18.7	18.1	16.2	14.9	13.0	11.9	12.1	13.3	14.5	15.8	17.5	(25)	
		MCWB	15.8	16.0	15.4	13.7	12.4	10.6	9.7	9.8	10.8	11.8	12.9	14.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.1	18.1	17.2	15.9	14.8	13.0	12.0	11.7	12.9	14.0	15.2	16.8	(27)	
		MCDB	19.9	19.8	19.0	17.2	16.1	14.1	13.1	13.0	14.8	15.9	17.2	18.6	(28)	
	2%	WB	17.2	17.4	16.6	15.1	14.0	12.2	11.1	11.1	12.2	13.2	14.4	16.2	(29)	
		MCDB	19.1	19.1	18.4	16.6	15.5	13.5	12.5	12.5	13.8	15.1	16.4	18.0	(30)	
	5%	WB	16.7	17.0	16.2	14.6	13.4	11.6	10.5	10.6	11.6	12.7	13.9	15.7	(31)	
		MCDB	18.5	18.6	18.0	16.2	14.9	13.1	12.0	12.1	13.3	14.5	15.8	17.5	(32)	
	10%	WB	16.2	16.5	15.7	14.1	12.7	11.0	10.0	10.1	11.1	12.2	13.4	15.2	(33)	
		MCDB	18.0	18.2	17.6	15.9	14.4	12.6	11.5	11.7	12.9	14.0	15.3	17.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.7	3.6	3.5	3.4	3.2	3.3	3.1	3.3	3.5	3.9	3.9	3.7	(35)	
		MCDBR	4.6	4.4	4.2	3.9	3.8	3.9	3.5	3.6	4.1	4.7	4.9	4.6	(36)	
	5% WB	MCWBR	2.8	2.8	2.8	2.8	2.8	3.2	3.0	2.8	2.9	3.2	3.0	2.9	(37)	
		MCWBR	3.9	3.9	3.5	3.3	2.9	3.1	2.9	3.0	3.5	3.8	4.1	3.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.341	0.335	0.319	0.311	0.291	0.288	0.287	0.300	0.320	0.328	0.333	0.335	(40)	
		taud	2.574	2.607	2.647	2.647	2.680	2.646	2.652	2.634	2.555	2.555	2.559	2.602	(41)	
	Ebn at Noon		986	967	940	880	836	800	829	880	922	962	988	998	(42)	
		Edh at Noon	103	95	84	72	59	56	59	71	90	99	104	102	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.61	5.86	4.58	3.04	2.06	1.55	1.78	2.60	3.79	5.14	6.33	6.59	(44)	
	RadStd		0.46	0.49	0.26	0.23	0.12	0.09	0.12	0.15	0.23	0.34	0.50	0.45	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	+0.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page